

Vibration-Generation Mechanism and Reduction Method in Linear Iron-Cored Permanent-Magnet Synchronous Motors at Stationary State

Eun Kyu Kim , Wontaek Song , Yoon Sik Kwon , Jae Hyun Kim, Hyeong Min Yoon ,
Hyo Geon Lee , and Jun Young Yoon 

Abstract—This article presents the theoretical studies and experimental validation of the vibration-generation mechanism and reduction method in linear iron-cored permanent-magnet (PM) motors at stationary state. While motor noise and vibration during moving operations have been extensively studied, such vibration issue at stationary state has been relatively less explored even though it can be critically problematic for high-precision applications where strict standstill of a motor is demanded. We identify a discrete position feedback in a servo control loop as a major culprit of the stationary-state motor vibration, and investigate the vibration-generation mechanism analytically and experimentally. Closed-loop dynamics related to such undesired vibration is analyzed by modeling a position-controlled linear iron-cored PM motor with a discrete encoder feedback. The analyzed model suggests that the dominant vibration frequency is determined by the control parameters regardless of the encoder quantization, while the corresponding vibration amplitude increases solely by the encoder quantization step size. Giving a standstill command to the linear motor stage testbed position controlled with different quantization step sizes, we have confirmed that the analyzed model consists of the experimental results as well as the simulated results. Understanding the vibration-generation mechanism of linear iron-cored PM motors, we also present a reduction method of such stationary-state motor vibration by a programmable-resolution incremental encoder. Using our method, we can reduce the motor vibration significantly while overcoming the speed limit of general quadrature encoders, providing a practical solution to various industrial applications

that require both high-throughput and high-precision performance.

Index Terms—Encoder quantization, motor vibration, regenerative control current, stationary-state vibration, vibration-generation mechanism.

I. INTRODUCTION

LINEAR motors are widely used in industrial applications as direct-drive actuators for rapid dynamic response and high-precision motion control [1]–[4]. There exist, however, various sources of noise and vibration that can impair servo performance of linear motors used in high-precision and high-throughput mechatronic systems [5]–[13] even without motor faults during operations [14]–[16].

Mechanisms of vibro-acoustic noise of linear motors can be different based on the operating states, which are mainly moving state and stationary state. The majority of prior-art on the motor noise studies the dominant sources of motor noise and vibration at moving states, such as force fluctuation and associated deformation of linear motor structures [10], [11]. Using Halbach magnet arrays [12], [13] and skewing either iron cores or magnets [17]–[19] are mostly used solutions that smoothen magneto-motive forces and thereby reducing the associated vibro-acoustic noise during moving operation.

The stationary-state motor vibration can be also critically problematic for industrial applications, such as automatic precision measurement systems, e.g., optical inspection systems [20]–[24], as shown in Fig. 1, where motor stages are required to be strictly standstill during the inspection process. For the motor vibration at stationary state, various disturbance sources can induce the vibration even if there are no electromagnetically driven force ripples. Vibrations from the system base, ground, or other devices, which are mechanically connected to the linear motor structure, can be the source of such stationary-state vibration [21]. In addition, high-voltage switching by the pulsewidth modulation (PWM) of motor drivers contains high-frequency harmonic components, which can be another source of linear motor vibrations at stationary state [25]. For these external vibration sources, vibration-isolating devices or noise shielding [26], [27] can be used as relatively simple and effective solutions. These solutions, however, require the usage and installation of additional devices and components.

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The authors are with the Department of Mechanical Engineering, Yonsei University, Seoul 03722, South Korea (e-mail: kek211@yonsei.ac.kr; wtsong@yonsei.ac.kr; dbstrakswo@gmail.com; jaehyun0924@yonsei.ac.kr; hyeongmy@yonsei.ac.kr; hyogeon007@yonsei.ac.kr; junyoung.yoon@yonsei.ac.kr).

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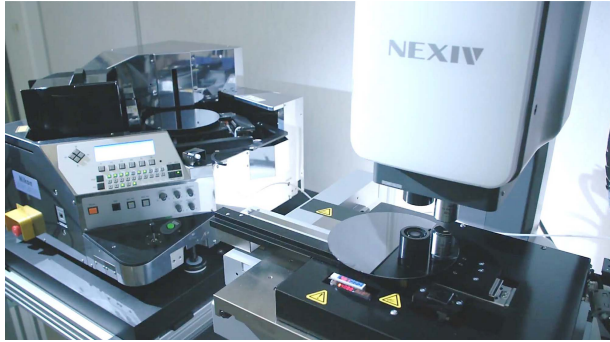


Fig. 1. Automated wafer inspection system, NEXIV VMZ-S3020 by Nikon [20], requiring strict standstill performance for high-precision inspection process.

The sensor quantization, on the other hand, could act as an internal disturbance source for the position-controlled servo systems to directly excite the motor stage, especially severely when passing through high-gain servo controllers [28]–[30]. It is yet difficult to find comprehensive studies and experimental investigation on the impact of such quantization issue on the motor noise and vibration at standstill command. We present the following in this article.

- 1) The analytic investigations on the effects of sensor quantization on the vibration of linear motors at stationary state.
- 2) Experimental validations of such stationary-state vibration-generation mechanism.
- 3) A practical vibration-reduction method with associated experimental demonstration.

The rest of this article is organized as follows. In Section II, we discuss the vibration-generation mechanism of a permanent-magnet linear synchronous motor (PMLSM) at stationary state in terms of encoder quantization and a high-gain servo controller. Section III presents the experimental testbed we designed and constructed to validate our hypotheses on the stationary-state motor vibration mechanism. We then present the associated experimental results and also demonstrate the practical method of reducing such motor noise in Section IV. Finally, Section V concludes this article.

II. VIBRATION-GENERATION MECHANISM OF PMLSM AT STATIONARY STATE

A. Quantization Error in Stationary Position

Position sensor measurements for a stationary target should be nominally constant. Analog signal values may, however, change in a small amount due to undesired disturbances, such as electrical noise from power electronics, electromagnetic interference, and disturbance forces, and vibrations from external environments. For such reasons, perturbations of analog sensor signals exist even when a servo system is commanded to stay stationary. Fig. 2(a) illustrates the output signal of a quantizer, such as an incremental encoder widely used in high-precision servo systems. Such a discretized signal remains unchanged if small perturbation of the analog signal is bounded in one

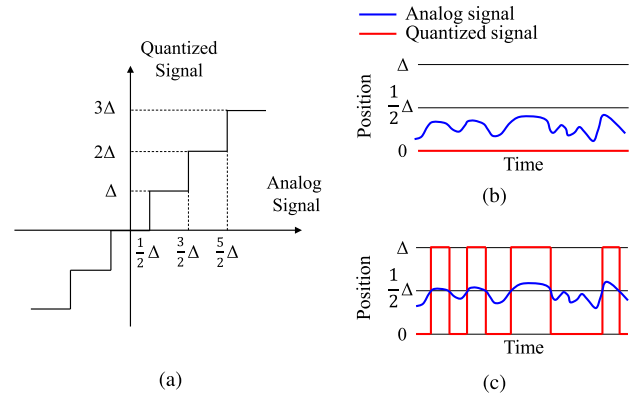


Fig. 2. Schematic of a quantizer in general. (a) It is assumed that the quantized signal switches its value at every mid-point. Δ represents the quantization step size. (b) Schematic of analog signal (blue curve), such as real position of a servo system, and its corresponding quantized signal (red line), e.g., incremental encoder position output signal at stationary state. Perturbation of the analog signal not contributing to the change in quantized output signal. (c) Perturbation of the analog signal adjacent to quantization threshold causing exaggerated and quantized change in the measured signal.

quantization step size, as shown in Fig. 2(b). When the analog signal happens to be adjacent to a quantization threshold, as shown in Fig. 2(c), however, even the same small perturbation in the analog signal can be read and in fact amplified by the magnitude of a sensor resolution as depicted by red pulses in the figure. Such exaggerated discrete position measurements generate unwanted control signals even at stationary, as they pass through high-gain servo controllers, thereby causing motor vibration at stationary state.

B. Regenerative Control Current by Quantized Feedback

In order to investigate the vibration mechanism at stationary state in detail, a control system simulation model of a PMLSM plant with a quantizer is built. Fig. 3(a) shows the block diagram of our simulation model, which consists of a linear controller $C(s)$, motor force constant K_t , motor plant $G(s)$, and a quantizer representing an incremental encoder with $\Delta = 1\mu\text{m}$ resolution. Here, we use a lead compensator for the controller, written as

$$C(s) = K \frac{\alpha\tau s + 1}{\tau s + 1} \quad (1)$$

where K , α , and τ represent a control gain, ratio of pole to zero, and time constant of the lead compensator, respectively. As for the motor plant, we use a simple mass system with a time delay of τ_d , which can be written as

$$G(s) = \frac{1}{ms^2} e^{-\tau_d s} \quad (2)$$

where m is the mass of a motor stage. As illustrated in the block diagram in Fig. 3(a), the linear motor plant $G(s)$ is exerted by the force F_{stage} , which consists of the motor thrust F_{thrust} and the disturbance force F_{PM} . Note that the disturbance force F_{PM} is caused by the magnetic saliency between the permanent magnets (PMs) and iron cores, as depicted in the figure, and it is considered to be constant over a small displacement of the

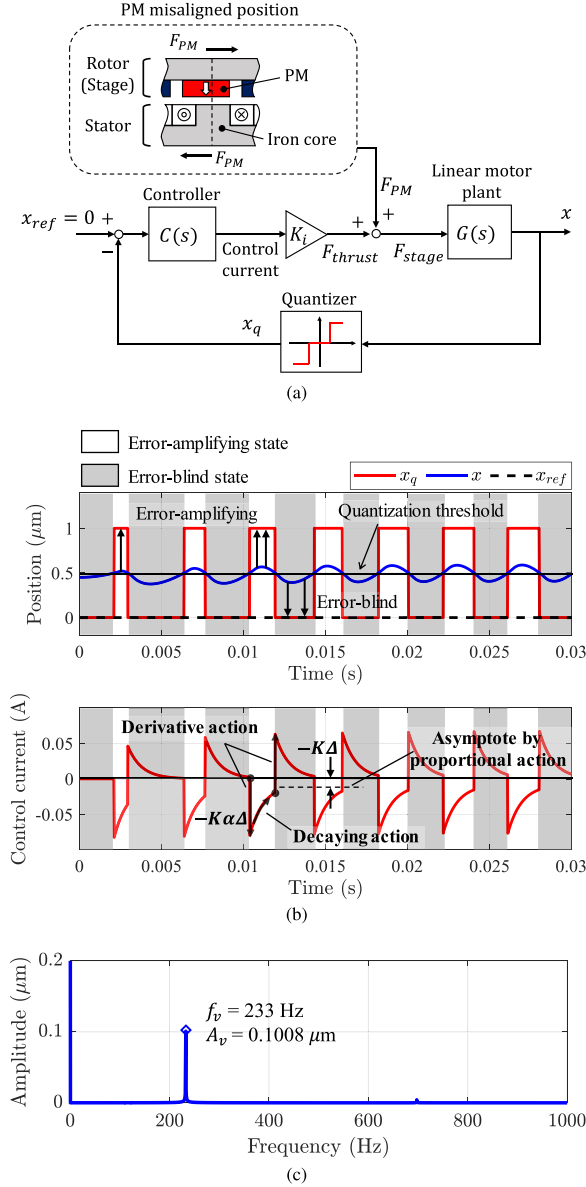


Fig. 3. Simulated closed-loop dynamics caused by quantized position error. (a) Block diagram of motor position control system. The variables x_q , x , x_{ref} , and K_i represent quantized position, true position, reference position, and the motor force constant of a linear motor, respectively, and F_{PM} denotes a constant disturbance force caused by the interaction between the permanent magnet (PM) and iron core. (b) Associated time-domain simulation results showing motor position and corresponding regenerative control current caused by the undesired quantization error at stationary state. (c) Frequency spectrum of motor vibration at stationary state obtained from the simulation results.

motor used in our simulation. Table I organizes the parameter values used for our simulations.

We implement the simulation with an initial position slightly below the quantization threshold of $\Delta/2$, and the corresponding results are plotted in Fig. 3(b). When the motor position x crosses over the quantization threshold $\Delta/2$, which we call the error-amplifying state, the position error changes from 0 to $-\Delta$, thereby inducing a negative-direction control current, as shown in the lower plot of Fig. 3(b). Such an induced control signal

TABLE I
MODEL PARAMETER VALUES USED IN SIMULATION

Symbol	Description	Value
K	Controller gain	8.72 A/mm
α	Lead compensator pole to zero ratio	10
τ	Lead compensator time constant	$8.388 \times 10^{-4} \text{ s}$
K_i	Thrust force constant of linear motor	87 N/A
m	Mass of linear motor stage	16 kg
τ_d	Time delay of linear motor plant	0.38 ms
Δ	Encoder resolution	1 μm
F_{PM}	Constant PM disturbance force	0.379 N

starts with its instantaneous peak generated by abrupt change of feedback signal x_q passing through the derivative action of the lead compensator (1). Its peak value can be calculated by the initial value theorem as

$$i(0) = \lim_{s \rightarrow \infty} s \frac{-\Delta}{s} K \frac{\alpha\tau s + 1}{\tau s + 1} = -K\alpha\Delta. \quad (3)$$

After the derivative action, the magnitude of control current decays exponentially with a time constant τ toward an asymptote determined by the proportional action of the controller. We can calculate such asymptote by the final value theorem as

$$i(\infty) = \lim_{s \rightarrow 0} s \frac{-\Delta}{s} K \frac{\alpha\tau s + 1}{\tau s + 1} = -K\Delta. \quad (4)$$

This control current stays negative until the quantized error signal becomes back to 0 as the motor position comes below the quantization threshold, which we call the error-blind state. At this error-blind state, we have the error changing from $-\Delta$ to 0, and this causes a positive peak control current with a magnitude of $K\alpha\Delta$ with the decaying lead compensator dynamics toward zero. This control current moves the motor position back to over the quantization threshold, making the position entering into the error-amplifying state again, and then the simulation model repeats the aforementioned process.

The magnitude of control current at the error-blind state is always less than that of the error-amplifying state. This is because the error-amplifying state has a nonzero decaying limit, which is the asymptote of $-K\Delta$ by the proportional control action. The error-blind state, on the other hand, has the zero decaying limit since the error signal is held zero. Such an unbalanced control current waveform can be balanced when a certain amount of constant disturbance force is added to the positive side of the control current, making the motor force excitation at both direction the same and thereby limit cycling the motor stage displacement. This is in fact the case for iron-cored PMLSMs where the magnetic saliency between the PMs and stator iron cores generates such a constant disturbance force, as illustrated in Fig. 3(a).

Fig. 3(c) shows the frequency spectrum of the regenerative vibration of the simulated motor position x , as depicted in the upper plot of Fig. 3(b). The frequency of $f_v = 233 \text{ Hz}$ and the amplitude of $A_v = 0.1008 \mu\text{m}$ are observed for the simulated motor vibration at a stationary state. The frequency and amplitude of such stationary-state vibration caused by the regenerative control current are also analyzed in detail in Section II-C, and the associated experimental measurements are presented in Section IV-A.

C. Analysis on Closed-Loop Dynamics by Quantized Error

When the constant disturbance force balances the motor excitation force between the error-amplifying state and the error-blind state, such regenerative control current depicted in the lower plot of Fig. 3(b) forces the motor to vibrate even at the stationary state. This occurs specifically when F_{PM} shown in Fig. 3(a) acts as an additional DC control current of $K\Delta/2$, that is

$$F_{PM} = K_i \frac{K\Delta}{2} \quad (5)$$

whose value is calculated to be 0.379 N in our experimental system discussed in Section III. The frequency f_v and the amplitude A_v of the regenerative vibration at the motor stationary state can be derived as follows. We consider the regenerative motor vibration to be a simple sinusoid, written as

$$x(t) = A_v \sin(\omega_v t) \quad (6)$$

with a steady-state vibration frequency of $\omega_v = 2\pi f_v$. We then take the first term of the Fourier series of the quantized square wave x_q shown in Fig. 3(b) to have

$$x_{q,1}(t) = \frac{2\Delta}{\pi} \sin(\omega_v t). \quad (7)$$

Comparing (6) and (7), we understand that the quantizer gives an additional gain of

$$K_v = \frac{2\Delta}{\pi A_v} \quad (8)$$

to the control loop shown in Fig. 3(a). Note that this additional gain, when greater than unity, in turn reduces the gain margin of the closed-loop system and might push the system onto the verge of instability. We calculate the frequency ω_v and the amplitude A_v of the regenerative vibration caused by such closed-loop system instability, by solving the characteristic equation of the closed loop shown in Fig. 3(a), written as

$$L(j\omega_v) = K_v C(j\omega_v) K_i G(j\omega_v) = -1. \quad (9)$$

In order to find ω_v , we use the phase relationship of (9) as

$$\angle L(j\omega_v) = \angle K_v C(j\omega_v) K_i G(j\omega_v) = (2n+1)\pi \quad (10)$$

for all integer number of n . Substituting (1) and (2), we can rewrite (10) as

$$\tan^{-1} \left(\frac{\tau\omega_v(\alpha+1)}{1-\alpha\tau^2\omega_v^2} \right) - \omega_v\tau_d = 2n\pi. \quad (11)$$

Since it is difficult to directly find the explicit solutions of ω_v from (11), we numerically obtain the set of ω_v solutions by interpolating the intersection points of the curves formed from the left- and right-hand sides of the equation with respect to ω_v . We then find the corresponding K_v value that makes the closed-loop system unstable for each ω_v by solving the magnitude relationship of (9) as

$$|L(j\omega_v)| = |K_v C(j\omega_v) K_i G(j\omega_v)| = 1. \quad (12)$$

Note that we can rearrange (12), using (1) and (8), to see the proportional relation of the vibration amplitude A_v to the control

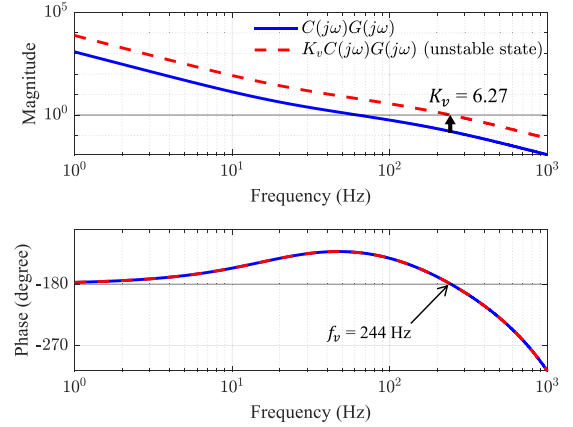


Fig. 4. Frequency response of the linear motor control system model showing the case without the quantization effect (blue solid curve) and an unstable case (red dashed line) with the additional gain K_v by the quantizer. Vibration frequency of 244 Hz is observed at the additional gain amplifier K_v of 6.27.

gain K and the quantization step Δ as

$$A_v = \frac{2\Delta K_i K}{\pi} \left| \frac{j\alpha\tau\omega_v + 1}{j\tau\omega_v + 1} G(j\omega_v) \right|. \quad (13)$$

The first set of solution $\omega_v = 1534$ rad/s and $K_v = 6.27$ gives the vibration frequency f_v of 244 Hz with the amplitude A_v of $0.1016 \mu\text{m}$. These values are close to the simulated results of 233 Hz and $0.1008 \mu\text{m}$, as shown in Fig. 3(c). The difference between the analytic solutions and the simulation results is due to the assumption we used in (7), where we considered the switching quantized signal during the regenerative vibration as only the fundamental sinusoid. Note that the subsequent solutions from (11) and (12) are negligible since the second solution gives $A_v = 1.08$ nm at $f_v = 2658$ Hz, which is below the resolution of our position measurement system as discussed in Section III.

Fig. 4 gives a graphical representation of (10) and (12) such that the closed-loop system becomes unstable (red dashed line) when the amplification effect (K_v) of the quantizer corresponds to the gain margin of the closed-loop system. Note that we can consider the regenerative vibration frequency ω_v as the phase-crossover frequency of the loop function $L(s)$. The phase-crossover frequency is in general placed at a higher frequency than the controller bandwidth, which indicates that the higher position controller bandwidth would lead to the higher regenerative vibration frequency ω_v . We can also see from (13) that the amplitude of the regenerative vibration A_v increases with the control gain K and the quantization step size Δ since the associated excitation current is proportional to the K and Δ , as shown in (3) and (4). Given that the control system parameters are maintained intact, we can conclude that the amplitude of the regenerative vibration due to the quantization effect and the associated regenerative control current increases solely with the quantization step size Δ .

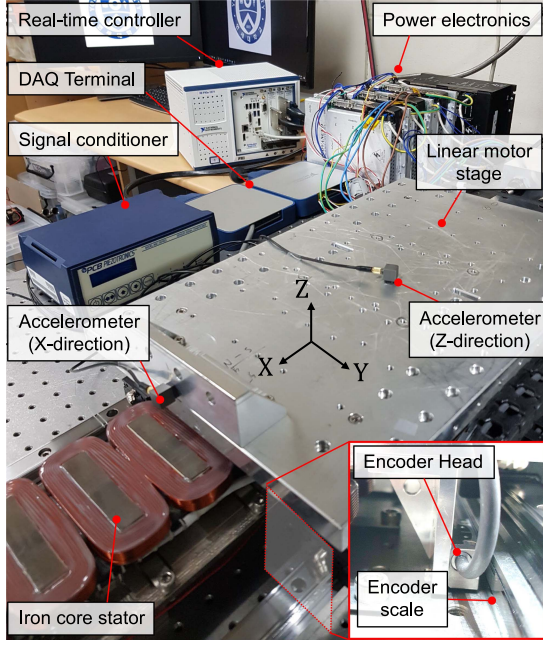


Fig. 5. Photograph of our experimental system setup. Linear motor stage is driven by an iron-cored PMLSM with its position measured in real-time by an optical encoder system. Encoder signal is used as the feedback to the servo control implemented in the real-time controller. Accelerometers are installed on the stage for vibration measurements.

III. EXPERIMENTAL TESTBED SETUP

A. Linear Motor System

Fig. 5 shows our linear motor stage testbed to experimentally validate the motor vibration-generation mechanism at stationary state. The stage is driven by a conventional three-phase PMLSM with an iron-cored armature as a stator and a PM track as a rotor. The PM track consists of eight poles with a 37.5-mm pole pitch, and makes a three-phase-four-magnet combination pair with the armature, which consists of six iron-cored windings with a 50-mm tooth pitch. The total mass of the moving stage is 16 kg, including the PM track. We measure the motor force constant of our PMLSM as 87 N/A. We use PWM power amplifiers (B30A40 by Advanced Motion Controls) to drive the motor phase coils. The necessary DC bus voltage to drive the motor coils are provided by a power supply unit (PS30A by AMC). We use an optical linear encoder system by Renishaw (T1031-50A for the read-head, RTL-C-S for the scale, and Ti0000A00 for analog signal interface) connected to our real-time controller (PXIe-8840 by National Instruments) so as to provide the real-time position feedback to our servo controller.

We design and implement our servo control loop, which runs at 10 kHz and has a -3 -dB bandwidth of 60 Hz to be able to command a stop to our linear stage testbed for the stationary-state motor vibration experiments. Accelerometers (320C33 by PCB) are used to measure the motor vibration at stationary state, as shown in Fig. 5. We use a signal conditioner (482C16 by PCB) to amplify the acceleration signals to achieve high signal-to-noise ratio. In the signal conditioner, the sensing range is adjusted to ± 9.8 m/s² by setting the sensor gain to 100

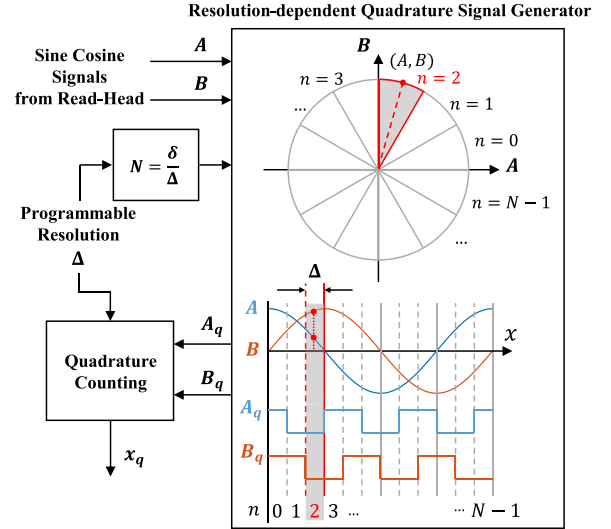


Fig. 6. Schematic of programmable-resolution incremental position measurement used in the feedback of the servo control loop. Sine and cosine encoder signals A, B are taken into our quadrature signal generator to locate the index n from the total number of quadrature sections N , which is determined by a programmable resolution Δ we define. According to the index n , digital quadrature signals of A_q and B_q with a 90° phase shift are generated. Such quadrature signals are used to calculate the incremental position measurement x_q . Note that the grating period of the encoder we use is $\delta = 20\mu\text{m}$.

for the accelerometers. A multifunction DAQ module (NI-6361 by NI) takes these amplified signals in the real-time control loop with the rate of 10 kHz, providing the Nyquist frequency of 5 kHz, which is sufficiently higher than the frequency range of our interest.

B. Programmable-Resolution Incremental Encoder

In order to validate the effect of quantization step size on the stationary-state vibration, we adjust the encoder resolution with our programmable quantizer implemented in our servo controller, allowing direct observations of the quantization effect without requiring any hardware changes in the position-sensing system. Fig. 6 shows the schematic of our programmable-resolution incremental encoder used for the vibration experiments discussed in Section IV. Depending on the programmable resolution Δ , the number of quadrature sections N per grating period of the encoder scale δ is determined as

$$N = \frac{\delta}{\Delta} \quad (14)$$

where N should be a multiple of 4 in order to generate the quadrature signals with full periods. Considering that the signals from a sinusoidal encoder read-head are nominally cosine and sine waveform with an amplitude of M , we can express the sampled encoder signals at a mechanical position of x as

$$A = M \cos \left(2\pi \frac{x}{\delta} \right) \quad (15)$$

and

$$B = M \sin \left(2\pi \frac{x}{\delta} \right). \quad (16)$$

The phase angle θ is obtained by using an arc-tangent function as

$$\theta = \text{atan2}\left(\frac{B}{A}\right), \quad \theta \in [0, 2\pi). \quad (17)$$

We can then find the index n of the quadrature section corresponding the current position x as

$$n = \text{floor}\left(N \frac{\theta}{2\pi}\right) \quad (18)$$

where the floor function rounds off the value calculated inside the parentheses. The digital quadrature signals A_q and B_q are generated according to the quadrature section index n as

$$A_q = \begin{cases} \text{HIGH}, & n(\text{MOD}4) = 0, 3 \\ \text{LOW}, & n(\text{MOD}4) = 1, 2 \end{cases} \quad (19)$$

$$B_q = \begin{cases} \text{HIGH}, & n(\text{MOD}4) = 0, 1 \\ \text{LOW}, & n(\text{MOD}4) = 2, 3 \end{cases} \quad (20)$$

where $n(\text{MOD}4)$ represents the division remainder of n by 4. The quadrature signals A_q and B_q obtained from (19) and (20) are 90° phase-shifted, as illustrated in Fig. 6, and we obtain the incremental position measurements x_q with conventional quadrature counting algorithms [31], but with a programmable resolution Δ that we define in real-time. Using this programmable-resolution encoder, we can adjust the encoder resolution on the fly without any hardware modification of our sensing system.

IV. EXPERIMENTAL RESULTS

A. Regenerative Vibration at Stationary State

We implement the servo control scheme shown in Fig. 3(a) with the parameters in Table I into our experimental testbed to conduct the motor vibration experiments at the stationary state. Fig. 7(a) shows the experimental results of such vibration with the motor position x (blue curve) and the quantized position x_q (red line) with a resolution step size of $1 \mu\text{m}$ defined by our programmable-resolution method. Note that the position x is measured with a high-resolution encoder to track the true displacement of the motor at the stationary states. When the position-controlled motor stage is commanded a stop, $x_{\text{ref}} = 0$, as shown with a black dashed line in the figure, and its position stays close to the quantization threshold of $0.5 \mu\text{m}$, as shown with a solid blue curve, small perturbations may cause occasional crossing of the quantization threshold. The position is then amplified to x_q , as shown with a solid red curve, resulting in abrupt changes in the position error and generating an regenerative control current, as shown in the lower plot of Fig. 7(a). Such regenerative current causes the motor vibration at stationary states, as discussed in Section II-B. Note that we observe in the experimental data of Fig. 7(a) the error-amplifying and error-blind states with the associated derivative and proportional control actions, as discussed with our simulation results illustrated in Fig. 3(b).

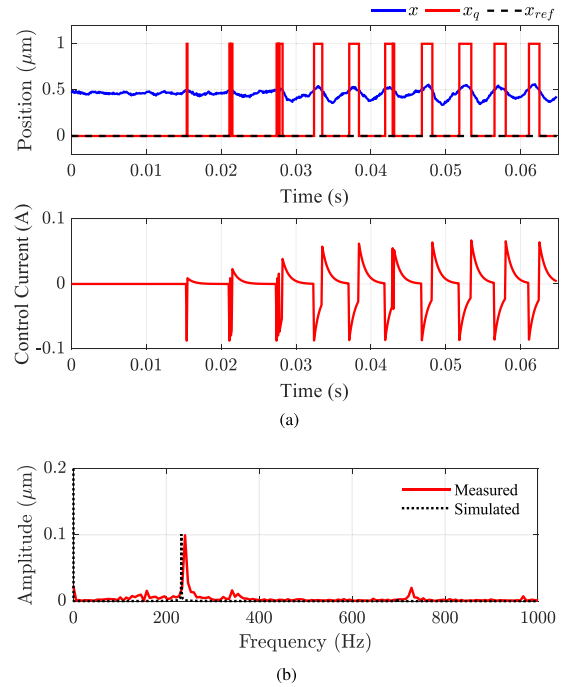


Fig. 7. Experimental measurements of the regenerative motor vibration at stationary state. (a) Position x , quantized and amplified position x_q with a resolution of $1 \mu\text{m}$, and corresponding control current showing regenerative behavior at stationary state. (b) Frequency spectrum of motor position x from the measurement data showing the vibration frequency f_v of 241 Hz and the amplitude A_v of $0.0993 \mu\text{m}$, which agree with the simulation results (black dotted curve) discussed in Section II.

We obtain the frequency spectrum of the measured motor position x , as shown in Fig. 7(b), to check the stationary-state vibration characteristics. Note that we use the original position signal x for the spectral analysis rather than the quantized signal x_q so as to observe the realistic vibrational behavior of the motor stage at the stationary state. The spectrum of the measured motor position shows the vibration frequency of $f_v = 241 \text{ Hz}$ with an amplitude of $A_v = 0.0993 \mu\text{m}$, which are consistent to the simulation-estimated values of 233 Hz and $0.1008 \mu\text{m}$, as depicted with a dotted black curve in Fig. 7(b). Such agreement manifests our understanding of the vibration-generation mechanism that the quantizer behaves as an additional gain amplifier, making the closed-loop stability marginal and thereby generating the vibration even at standstill commands. As analytically predicted in Section II-C, such stationary-state vibration of PMLSMs is heavily dependent on the quantization step size. We present the associated experimental results in the following section.

B. Effect of Quantization Step Size

In order to demonstrate the effect of quantization step size on the stationary-state motor vibration, we experimentally measure the motor position at the standstill command, with different sensor quantization step sizes of $\Delta = 2.5, 1, 0.5$, and $0.1 \mu\text{m}$ set by our programmable quantizer discussed in Section III-B. Fig. 8 shows the frequency spectra of measured motor position data x with different encoder resolutions. The dominant vibration

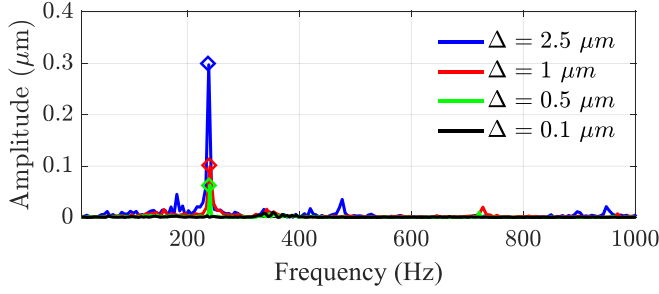


Fig. 8. Frequency spectra of the motor position x experimentally measured at stationary state with different position feedback resolutions. As predicted in Section II, the dominant vibration frequencies remain constant regardless of the encoder resolution while the vibration amplitudes show significant dependency on the encoder resolution value. Note that the control current induced by the quantized position signal of $\Delta = 0.1 \mu\text{m}$ (black line) is no longer sufficient to generate observable motor vibration at stationary state.

TABLE II
REGENERATIVE VIBRATION FREQUENCY f_v

Δ	Modeled	Simulated	Measured
$2.5 \mu\text{m}$	244 Hz	233 Hz	238 Hz
$1 \mu\text{m}$	244 Hz	233 Hz	241 Hz
$0.5 \mu\text{m}$	244 Hz	233 Hz	240 Hz
$0.1 \mu\text{m}$	244 Hz	233 Hz	-

TABLE III
REGENERATIVE VIBRATION AMPLITUDE A_v

Δ	Modeled	Simulated	Measured
$2.5 \mu\text{m}$	$0.2539 \mu\text{m}$	$0.2519 \mu\text{m}$	$0.2971 \mu\text{m}$
$1 \mu\text{m}$	$0.1016 \mu\text{m}$	$0.1008 \mu\text{m}$	$0.0993 \mu\text{m}$
$0.5 \mu\text{m}$	$0.0508 \mu\text{m}$	$0.0504 \mu\text{m}$	$0.0635 \mu\text{m}$
$0.1 \mu\text{m}$	$0.0102 \mu\text{m}$	$0.0101 \mu\text{m}$	-

frequency f_v and amplitude A_v are organized in Tables II and III, comparing with the analytical model and simulation results for all cases.

We observe from the experimental results in Fig. 8 that the measured vibration frequency f_v stays almost the same regardless of the sensor resolution change. This agrees with our prediction in (11) with ω_v showing no dependency on the resolution size, as discussed in Section II-C. We also see that the amplitude A_v of the regenerative stationary-state vibration increases almost proportionally with the quantization step size, as predicted by (8) and (12). As can be seen from Tables II and III, the values of experimentally measured frequencies f_v and amplitudes A_v are close to those of the analytical model, and the simulated results are discussed in Section II. The small discrepancies between the measured data and modeled values come from the modelling assumption we use in (7), simplifying the quantized signal as sinusoidal. Note that the experimental data for $\Delta = 0.1 \mu\text{m}$ are omitted in the tables since the vibration amplitude is insufficient to trigger the stage motion, as shown in Fig. 8. In this case, we also speculate that the microscale nonlinear stiffness and damping of the linear guide friction in the presliding regime [32] provide sufficient resistance to the disturbance force by the encoder error signal at the stationary state, thereby suppressing the regenerative vibration.

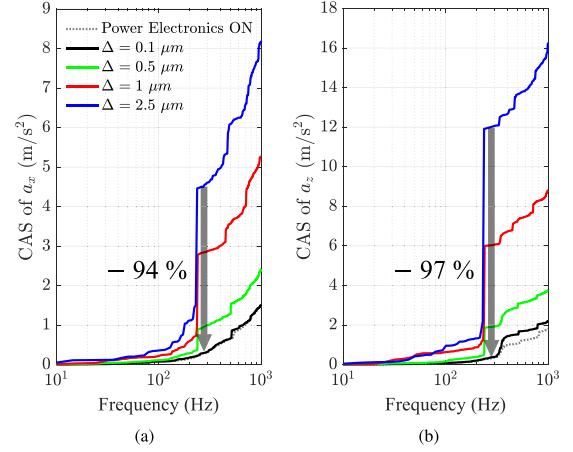


Fig. 9. CAS of the experimentally measured motor vibration caused by regenerative control current with different sensor quantization step sizes. Vibration measurement in the case of power electronics turned ON with the controller OFF is plotted (gray dotted curves) as a reference. Frequency with the steepest rise in CAS indicates the major vibration component, corresponding to the regenerative vibration frequency at 240 Hz. (a) x -direction vibration measurements in m/s^2 , showing 94 % reduction by changing the resolution step size from 2.5 to $0.1 \mu\text{m}$. (b) z -direction measurements in m/s^2 , showing 97 % reduction by using a smaller resolution of $0.1 \mu\text{m}$.

The stationary-state motor vibration is also measured by the accelerometers mounted on the stage both in the x - and z -directions, as shown in Fig. 5. The cumulative amplitude spectra (CAS) of the measured motor vibration data in m/s^2 are plotted in Fig. 9 with different sensor quantization step sizes of $\Delta = 2.5, 1, 0.5$, and $0.1 \mu\text{m}$. We also observe from the CAS plots that the regenerative vibration frequency remains the same regardless of the encoder resolution showing a steep increase at around 240 Hz for all cases. The vibration amplitude, on the other hand, decreases significantly with smaller encoder resolution sizes, showing a smaller step in the measured CAS. Using a finer encoder resolution of $0.1 \mu\text{m}$, we achieve a significant reduction of the stationary-state motor vibration by 94 % and 97 % in m/s^2 for the x - and z -directions, respectively, compared with the case of $2.5 \mu\text{m}$ resolution. Note that the slight rise in CAS of a_z at around 360 Hz, as shown in Fig. 9(b), is due to the first flexible mode of our motor stage, which is excited by the PWM switching noise from the motor driver.

C. Vibration Reduction by Programmable-Resolution Encoder

Fig. 10(a) shows the experimental results of the position trajectory following of the motor stage, which is position-controlled with a different incremental resolution of $\Delta = 1 \mu\text{m}$ and $\Delta = 0.1 \mu\text{m}$. The motor stage is commanded a stop after following a fourth-order polynomial reference trajectory with $a_{\max} = 750 \text{ mm/s}^2$, $v_{\max} = 300 \text{ mm/s}$, and a travel length of 200 mm. When the $1 \mu\text{m}$ resolution feedback is used, the blue line in Fig. 10(a), the motor tracks the given trajectory during the nonzero traveling range. After about 1.27 s, when the standstill command starts, however, the regenerative motor vibration is induced due to the effect of the quantized position

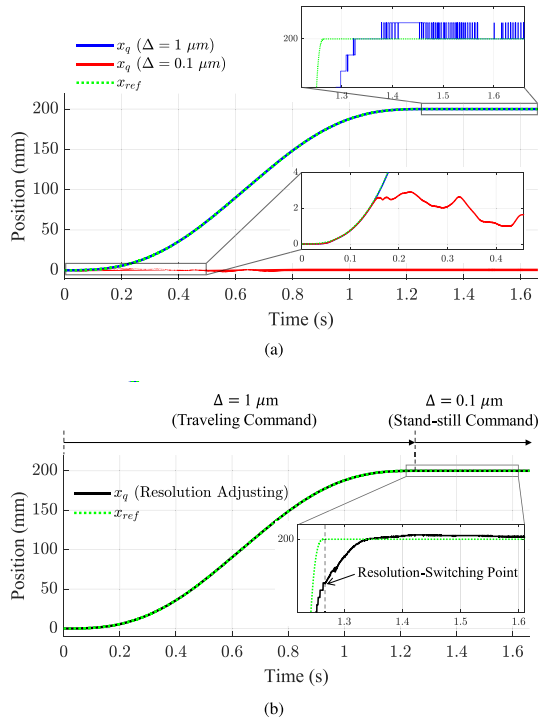


Fig. 10. Experimental data of fourth-order polynomial position trajectory profile control. After traveling, the reference trajectory is set to the constant set point for a standstill command to the motor stage. (a) Position-controlled trajectory performance with $1\ \mu\text{m}$ resolution (blue line) feedback. Regenerative vibration of motor position x_q with quantization step size ($\Delta = 1\ \mu\text{m}$) is observed during standstill command. With $0.1\ \mu\text{m}$ resolution (red line) feedback, position tracking fails due to the operating velocity limit. (b) Resolution is adjusted in real-time for the linear motor stage to travel with $1\ \mu\text{m}$ resolution feedback for nonzero velocity operation while using $0.1\ \mu\text{m}$ resolution feedback for standstill command. Regenerative vibration of motor position x_q due to the quantization step size is significantly reduced compared with the $1\ \mu\text{m}$ resolution case in (a) as shown in the magnified view.

feedback signal, as shown in the magnified view in the figure. While a finer resolution of $0.1\ \mu\text{m}$ could be used [see red line in Fig. 10(a)], in order to reduce the unwanted stationary-state vibration, the encoder misses the quadrature signal counting due to its operating speed limit, thereby failing the motor trajectory following.

In order to overcome such tradeoff between the encoder resolution and the decoding operational speed limit, we adjust the encoder resolution in real-time depending on the operating mode of our linear stage, as shown in Fig. 11, by utilizing the programmable-resolution incremental encoder method discussed in Section III-B. We use a fine encoder resolution, $\Delta_{\text{standstill}} = 0.1\ \mu\text{m}$, when the motor is commanded to maintain its stationary position so as to reduce the stationary-state vibration. For traveling operations, on the other hand, we switch to a relatively coarse resolution, $\Delta_{\text{traveling}} = 1\ \mu\text{m}$, to avoid the encoder speed limit so that the position-controlled motor stage can meet faster travel demands. Fig. 10(b) shows the experimental results applying our encoder-resolution adjusting algorithm based on the motor operation state. The encoder resolution is programmed to be $1\ \mu\text{m}$ while the motor follows the given nonzero trajectory, and is switched to $0.1\ \mu\text{m}$ from the beginning

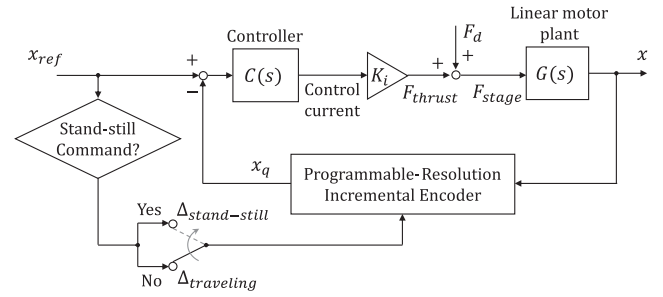


Fig. 11. Block diagram of the proposed vibration-reduction method utilizing the programmable-resolution incremental encoder. Depending on the motor operation mode, the encoder resolution is adjusted in real-time. The fine resolution $\Delta_{\text{stand-still}}$ is used to reduce the stationary-state vibration for standstill commands, while the coarse resolution $\Delta_{\text{traveling}}$ is selected to overcome the decoding operational speed limit during traveling operations.

of the standstill command. Using our method, we significantly reduce the stationary-state motor vibration without any hardware addition or modification and also without sacrificing the operating speed of the motor stage.

V. CONCLUSION

In this article, we investigate analytically and experimentally the vibration-generating mechanism of an iron-cored PMLSM at the stationary state, and demonstrate the vibration-reduction method. We discuss the effects of sensor quantization as a root cause of undesired vibration in a position-controlled PMLSM at standstill commands. The motor position can be quantized and exaggerated by an incremental encoder, and when it is utilized as a feedback signal in a servo control loop, such discrete measurement acts as an additional gain amplifier, causing regenerative current commands and associated vibration.

Based on the understandings of such vibration-generating mechanism, we propose a vibration-reduction method utilizing our programmable-resolution encoder and achieve a reduction of the stationary-state motor vibration by more than 94% in m/s^2 . We also simultaneously keep the motor operation-speed capability while reducing the stationary-state motor vibration, utilizing our programmable-resolution incremental encoder in real-time.

The motor vibration issue at stationary state has been relatively less explored even though it can be critically problematic for industrial applications, such as automatic inspection systems where the strict standstill of motor stages is required. We discuss the comprehensive studies of the vibration-generating mechanism at motor stationary states, which can provide practical solutions for reducing stationary-state vibrations in precision engineering applications that require both high-throughput and high-precision performance. We are currently conducting studies, as our future work, on the moving-state vibrations of our PMLSM stage used in this article, and also investigating the simultaneous vibration-reduction methods both at the stationary and moving states of the PMLSM.

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Eun Kyu Kim received the B.S. degree in 2020 in mechanical engineering from Yonsei University, Seoul, South Korea, where he is currently working toward the Ph.D. degree in mechanical engineering.

His research interests include design and precision motion control of electromagnetic and electromechanical systems.



Wontaek Song received the B.S. degree in mechanical engineering from Chungnam National University, Daejeon, South Korea, in 2019. He is currently working toward the Ph.D. degree in mechanical engineering with Yonsei University, Seoul, South Korea.

His research interests include modeling and precision motion control of servo systems, and modeling and optimization for energy consumption reduction of machine tool.



Yoon Sik Kwon received the B.S. degree in 2020 in mechanical engineering from Yonsei University, Seoul, South Korea, where he is currently working toward the M.S. degree in mechanical engineering.

His research interests include manufacturing mechatronics and electromagnetic machine design, and precision motion control.



Jae Hyun Kim received the B.S. degree in 2021 in mechanical engineering from Yonsei University, Seoul, South Korea, where he is currently working toward the Ph.D. degree in mechanical engineering.

His research interests include analytical modeling and design optimization of electromagnetic machines.



Hyeong Min Yoon received the B.S. degree in 2021 in mechanical engineering from Yonsei University, Seoul, South Korea, where he is currently working toward the Ph.D. degree in mechanical engineering.

His research focuses on the design and precision motion control of multidegrees of freedom actuators.



Hyo Geon Lee received the B.S. degree in 2021 in mechanical engineering from Yonsei University, Seoul, South Korea, where he is currently working toward the Ph.D. degree in mechanical engineering.

His research interests include electromagnetic actuator design and precision motion control.



Jun Young Yoon received the B.S. degree from Yonsei University, Seoul, South Korea, in 2009, and the M.S. and Ph.D. degrees from the Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, in 2011 and 2017, respectively, all in mechanical engineering.

He is currently an Assistant Professor with the Department of Mechanical Engineering, Yonsei University. From 2017 to 2019, he was a Postdoctoral Researcher of mechanical engineering and biological engineering with MIT. His research interests include manufacturing mechatronics and robotics system design, electromagnetic and electromechanical machine design, mechatronic devices for biomedical applications, and precision motion control.